

Replicating Table 2, Panel B (5-Fold) from Chernozhukov et al. (2018)

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May 31, 2026

1 The Original Paper

This study replicates a specific result from *Double/Debiased Machine Learning for Treatment and Structural Parameters* by Victor Chernozhukov, Denis Chetverikov, Mert Demirer, Esther Duflo, Christian Hansen, Whitney Newey, and James Robins, published in *The Econometrics Journal* in 2018 (Chernozhukov et al., 2018).

The paper addresses a fundamental problem in causal inference: when the number of controls is large relative to the sample size, naïvely plugging machine learning estimates of nuisance functions into a structural equation produces estimators that are biased at the $\sqrt{N}$ rate due to regularization bias and overfitting. The authors show that this bias can be removed by combining two ingredients: (1) *Neyman-orthogonal* moment conditions, which have reduced sensitivity to errors in nuisance estimation, and (2) *cross-fitting*, an efficient form of sample-splitting that prevents overfitting from contaminating the structural estimate. The resulting procedure is called Double/Debiased Machine Learning (DML).

2 The Result of Interest

This replication targets the 5-fold cross-fitting row of Panel B (Partially Linear Regression model) of Table 2, which reports the estimated Average Treatment Effect (ATE) of 401(k) eligibility (`e401`) on net total financial assets (`net_tfa`) across six machine learning methods and an ensemble. Panel A of Table 1 reproduces the paper’s targets exactly.

This particular result is the most directly interpretable in the paper: it translates the DML procedure into a concrete, policy-relevant estimate using a well-known dataset, and directly demonstrates that the choice of nuisance learner has only modest effects on the structural parameter once debiasing is applied correctly.

3 Data and Methods

Data

The data come from the 1991 Survey of Income and Program Participation (SIPP), following the sample construction of Abadie (2003). The raw file `input/sipp1991.dta` is loaded by `code/preprocess.R`, which selects the eleven analysis variables and drops incomplete cases, producing a balanced sample of household-level observations saved to `temp/clean_data.rds`. The outcome variable is net total financial assets (`net_tfa`) and the treatment indicator is whether the household’s employer offers a 401(k) plan (`e401`). Nine covariates are included: age, income, education, family size, marital status, two-earner status, defined benefit pension status, IRA participation, and home ownership.

Methods

The partially linear regression (PLR) model is:

$$Y = D\theta_0 + g_0(X) + U, \quad E[U \mid X, D] = 0, \quad (1)$$

$$D = m_0(X) + V, \quad E[V \mid X] = 0, \quad (2)$$

where Y is net financial assets, D is 401(k) eligibility, X is the vector of controls, and θ_0 is the ATE of interest. The nuisance functions g_0 and m_0 are estimated via cross-fitting with $K = 5$ folds. Six learners are used for nuisance estimation: Lasso (with a rich polynomial and interaction feature matrix matching the original paper), Regression Tree, Random Forest, Boosting, Neural Network, and an Ensemble average. The DML2 score function is used throughout. Results are aggregated across 100 random sample splits, reporting the median ATE and two standard error measures.

4 Replication Results

Table 1 presents the replicated results alongside the paper’s targets. In plain terms, the ATEs indicate that 401(k) eligibility increases net financial assets by roughly \$8,000–\$9,700 depending on the learner used for nuisance estimation, a large and economically meaningful effect relative to the sample mean of \$18,052. The stability of this estimate across very different machine learning methods, from a penalised linear model to a neural network, directly illustrates the DML estimator’s central claim of robustness to nuisance function specification.

Table 1: Replication of Table 2, Panel B (5-Fold): Estimated ATE of 401(k) Eligibility on Net Financial Assets (Partially Linear DML).

	Lasso	Reg. Tree	Forest	Boosting	Neural Net	Ensemble	Best
<i>Panel A: Paper targets (Chernozhukov et al., 2018)</i>							
ATE	8187	8871	9247	9110	9038	9166	9215
	[1558]	[1418]	[1328]	[1328]	[1355]	[1310]	[1312]
	(1298)	(1358)	(1295)	(1314)	(1322)	(1299)	(1294)
<i>Panel B: This replication</i>							
ATE	9676	9026	9269	8983	9213	9338	9667
	[1365]	[1394]	[1328]	[1331]	[1649]	[1330]	[1366]
	(1347)	(1356)	(1308)	(1325)	(1495)	(1311)	(1345)

The replication closely matches the paper for tree-based and ensemble methods: Random Forest (9,269 vs. 9,247), Boosting (8,983 vs. 9,110), Ensemble (9,338 vs. 9,166), and Best (9,667 vs. 9,215) all fall within a few hundred units of their targets, well within one standard error. The Neural Network ATE of 9,213 modestly exceeds the paper’s 9,038 by approximately 175 units, likely due to random variation in weight initialisation across splits. The Lasso ATE of 9,676 exceeds the paper’s 8,187 by approximately 1,500, likely reflecting the reduced polynomial degree used here (`poly(inc, 4)` versus `poly(inc, 8)` in the original). The two SE rows capture different uncertainty sources: the bracket row $[\cdot]$ adds split-to-split variability on top of the within-split SE reported in the parenthesis row $(\cdot)$; both are closely matched across all methods, confirming the score function and aggregation are correctly implemented.

5 What I Learned

The most straightforward aspect of the replication was the DML estimator itself. Once the cross-fitting loop and score function are correctly coded, the structural estimates fall out cleanly and the aggregation across 100 splits is mechanical. The framework’s robustness to learner choice, its central theoretical claim, is also directly visible in the results: once nuisance functions are estimated reasonably well, θ_0 is stable across very different ML methods, confirming the paper’s key insight empirically.

The hardest part was matching the nuisance function specification. The paper’s original R code contains a compact one-line formula (`x1`) for the Lasso feature matrix that expands to hundreds of columns of polynomials and pairwise interactions, but this detail does not appear in the paper text at all; it only appears in the code. Without reading `401K.R` carefully, it

would be easy to run Lasso on the raw nine covariates and obtain a badly biased estimate, incorrectly concluding the replication failed. Computational cost was a secondary challenge: with over 5,000 total ML fits, runtime exceeded seven hours until Forest and GBM settings were tuned. Exact seed matching with the original is also not possible since the paper uses parallel execution with per-worker seeds. Both issues point to the same lesson: code-level documentation matters as much as methodological description. If I were the original author, I would explicitly state the feature construction for penalised learners and the parallelisation setup in a replication appendix.

References

- Abadie, A. (2003). Semiparametric instrumental variable estimation of treatment response models. *Journal of Econometrics*, 113(2):231–263.
- Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., and Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, 21(1):C1–C68.